LECTURE 2

GEOSPATIAL DATA

What is geodata?

PRESS → OR SPACE.

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UNIVERSITÄT BONN

Learning Outcomes

Lecture

- Understand and compare major geospatial data sources from GIS, remote sensing, meteorology, geophysics, and environmental monitoring.
- Distinguish common geospatial data modalities, including raster images, vector layers, point data, point clouds, time series, trajectories, and gridded fields.
- Explain key properties of geospatial data, including spatial resolution, temporal resolution, spectral resolution, coverage, uncertainty, and coordinate reference systems.
- Describe common geospatial data formats and access patterns used in Python-based workflows.
- Match geospatial data sources and modalities to suitable environmental and socio-ecological applications.
- Identify practical challenges such as scale mismatch, missing data, sampling bias, spatial autocorrelation, and heterogeneous data quality.

Lab

- Access and preprocess raster, vector, and time-series geospatial data using Python-based tools.
- Combine multiple geospatial data layers into a common spatial reference and resolution.
- Extract training samples or analysis regions from geospatial datasets.
- Evaluate practical limitations of data quality, coverage, scale, and interoperability.

Coordinates

U.S. Government — Extracted from the PDF version of the ***Vol 26-4 2004 DISAM Journal*** (direct PDF [1]), via **Wikipedia**.

Coordinates and Location

How many coordinates do we need to uniquely define a spatio-temporal location?

3 spatial dimensions

Cartesian (x, y, z) or
Spherical/Ellipsoidal (λ, φ, r) or *or UTM Projections (Easting, Northing, Height)*

Where is it in space?

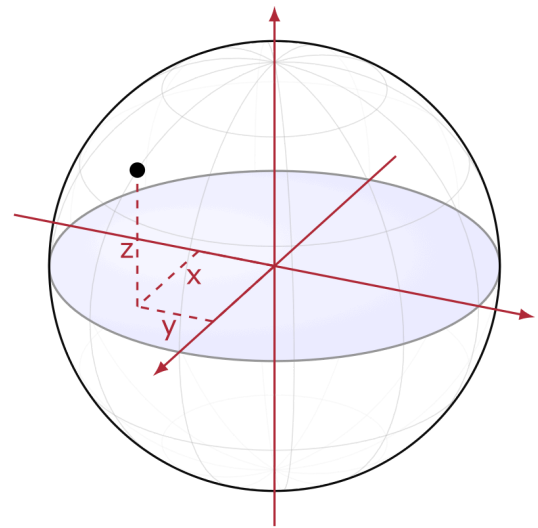
1 temporal dimension

t

When does it occur?

Cartesian Coordinates x,y,z

A location can be represented by three orthogonal coordinates $\mathbf{p} = (x, y, z)$ in an Earth-Centered, Earth-Fixed (ECEF) coordinate system



Spherical Coordinates

Expressing location on the surface of Earth is often more practical than working directly in 3D Cartesian coordinates $x, y, z \mapsto \lambda, \varphi, r$ of longitude λ, latitude φ, and constant radius r

Spherical to Cartesian

$$x = r \cos\varphi \cos\lambda$$

$$y = r \cos\varphi \sin\lambda$$

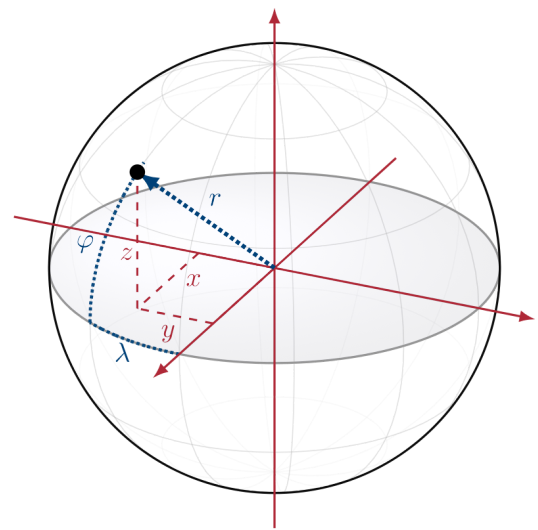
$$z = r \sin\varphi$$

Cartesian to spherical

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\lambda = \operatorname{atan2}(y, x)$$

$$\varphi = \arcsin\left(\frac{z}{r}\right)$$



Ellipsoidal Coordinates

But the Earth is not a sphere: It is an ellipsoid.

Geodetic to Cartesian

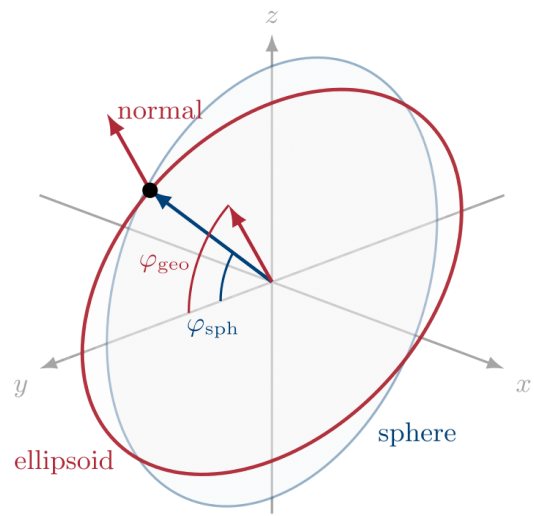
$$
\begin{aligned}
x &= (N(\varphi) + h)\cos\varphi\cos\lambda \\
y &= (N(\varphi) + h)\cos\varphi\sin\lambda \\
z &= \left((1-e^2)N(\varphi) + h\right)\sin\varphi
\end{aligned}
$$

$$
N(\varphi) = \frac{a}{\sqrt{1 - e^2\sin^2\varphi}}
$$

$$
e^2 = \frac{a^2 - b^2}{a^2}
$$

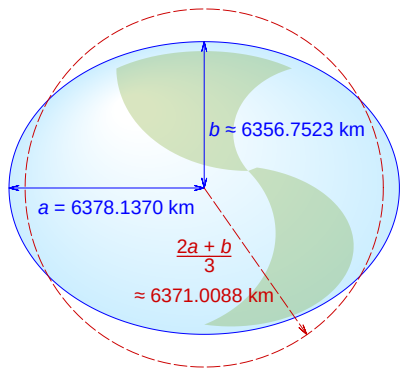
Cartesian to Geodetic

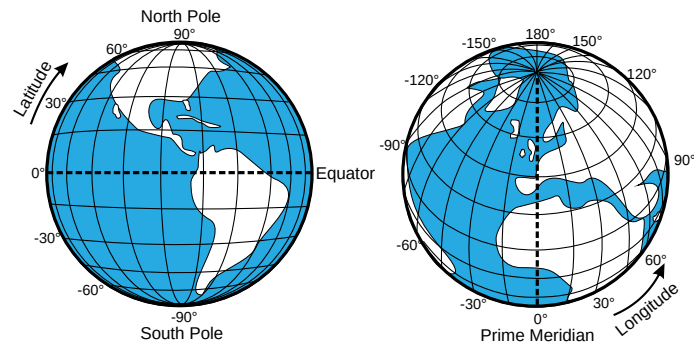
The inverse is usually computed iteratively.
See Wikipedia .



World Geodetic System 1984 (WGS 84)

WGS 84 is the global reference system used by GPS and most web mapping workflows.

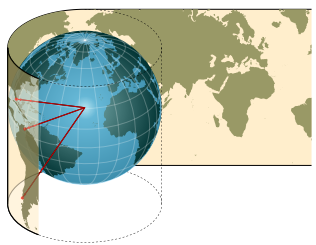




Cylindrical Map Projections and Web Mercator

Cylindrical projections - Mercator

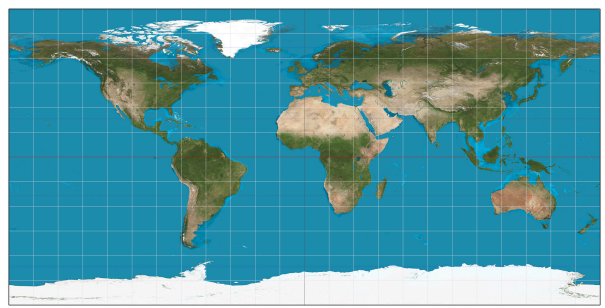
A cylinder touches or cuts the globe along a line or lines. The classic case is a cylinder around the equator.



Cylindrical projections minimize distortion along their standard line(s).

Web Mercator

Web Mercator is the projection used by many web maps. It is a cylindrical, conformal projection: local shapes are preserved, but areas grow strongly toward the poles.



Web maps trade area accuracy for visually stable local shapes and simple tiled rendering.

Cylindrical Map Projections and UTM

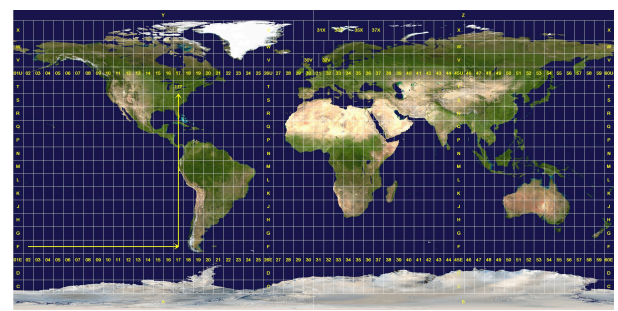
Cylindrical projections - UTM

The Universal Transverse Mercator system rotates the cylinder: each zone uses a transverse cylinder around a local central meridian.

UTM turns longitude/latitude into local metric coordinates: easting and northing.

Universal Transverse Mercator (UTM)

Pre-defined UTM zones give geodata local coordinate systems with comparatively low distortion inside each zone.



UTM uses different cylinders to avoid local distortions.

Radius: Atmosphere and Ocean

Geospatial fields are not only horizontal.

They also vary vertically:

- wind fields change across atmospheric columns
- ocean currents differ at the surface and at depth
- temperature, humidity, pressure, and salinity depend on height or depth
- many Earth-system processes require a vertical coordinate

Location often means latitude, longitude, and height or depth.

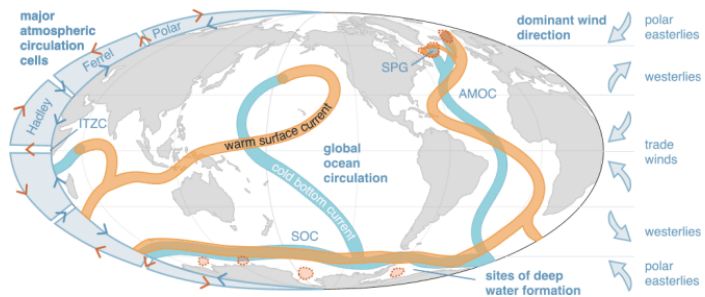


Figure: University of Exeter, ***Atmospheric circulation cells, dominant wind directions, key ocean basins, surface currents and deep water formation sites***. Licensed under CC-BY-SA: Creative Commons Attribution-Share Alike 4.0 International. The rest of this slide deck remains CC-BY-NC.

Time: The Earth is Dynamic

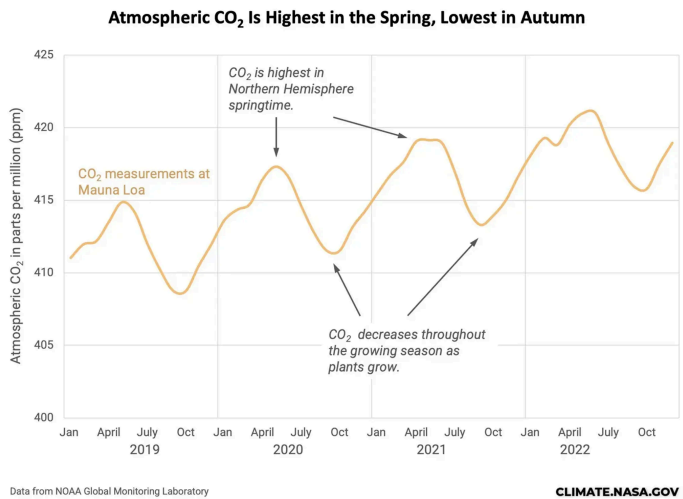


Figure: NASA Science, **Carbon Dioxide**.

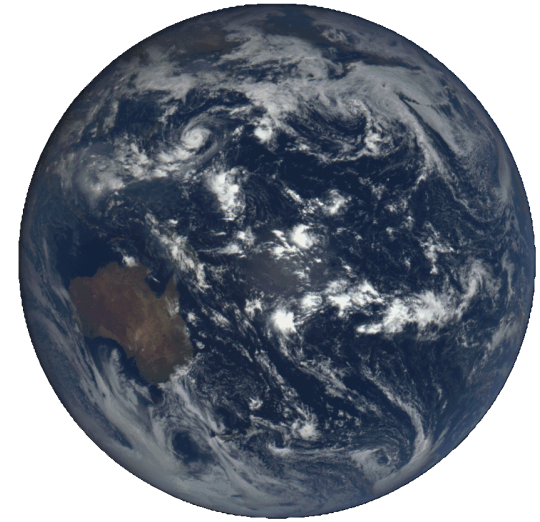
Figure: Google Earth Engine, **MODIS NDVI time series animation**. Licensed under CC-BY.

Time: Common Frequencies

Earth observations contain processes at many characteristic temporal frequencies.

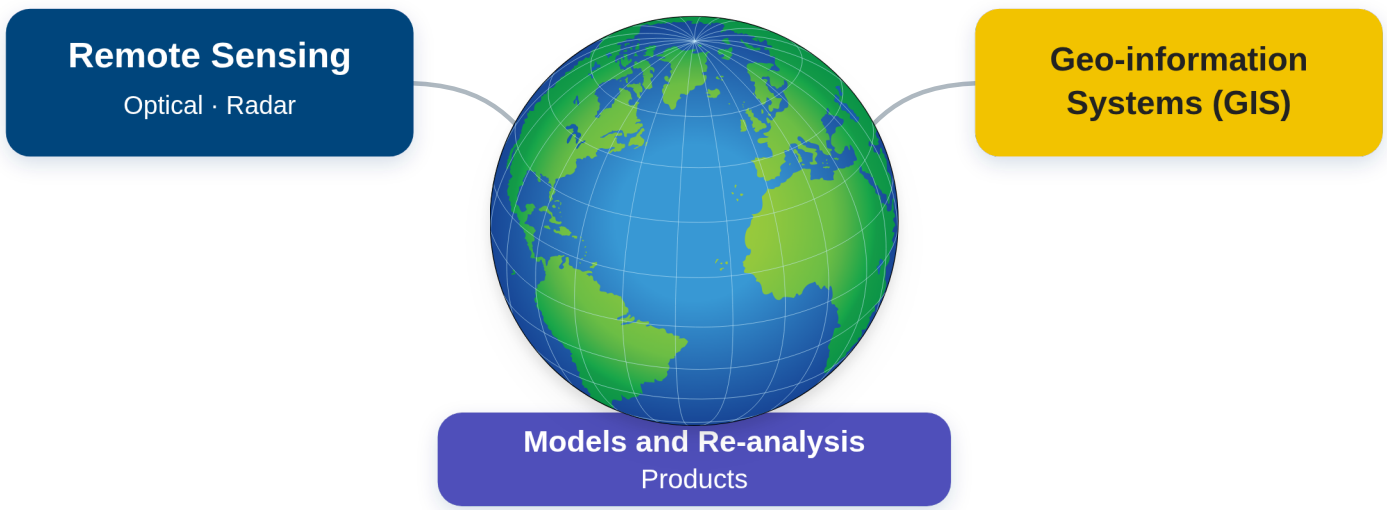
Time scale	Frequency	Examples
Seconds to minutes	High frequency	Wind gusts, turbulence, lightning, traffic, waves, sensor noise
Minutes to hours	Sub-daily	Cloud motion, precipitation cells, tides, urban mobility, river discharge
~12 hours	Semi-diurnal	Ocean tides, coastal water levels
24 hours	Daily / diurnal	Temperature cycle, solar radiation, human activity, vegetation
Several days	Weather	Storms, pressure systems, cloud regimes, heatwaves, cold fronts
Weekly	Anthropogenic	Commuting, energy use, shipping patterns, some air pollution signals
Monthly / ~29.5 days	Lunar	Spring-neap tide cycle, moon-related illumination effects
Seasonal / annual	Yearly	Phenology, crop cycles, snow cover, monsoon, sea ice, temperature
Interannual	2-7 years	ENSO / El Nino, drought cycles, vegetation anomalies
Decadal	10+ years	Climate variability, land-use change, glacier retreat, urban expansion
Multi-decadal to centennial	Long-term trend	Climate change, sea-level rise, ecosystem shifts

Remote Sensing



Campbell & Shin (2011), Figure 1.8.

Overview Section - Data Sources



Overview Section - Data Formats

Points
coordinate → values

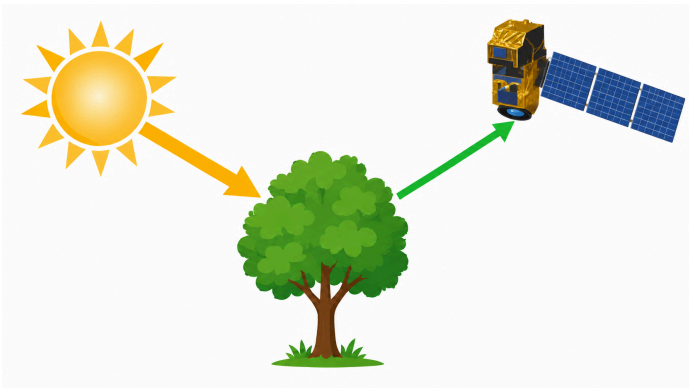
Lines
coordinate list → values

Polygon
closed coordinate list → values

Raster
coordinate grid → values

Active and Passive Remote Sensing

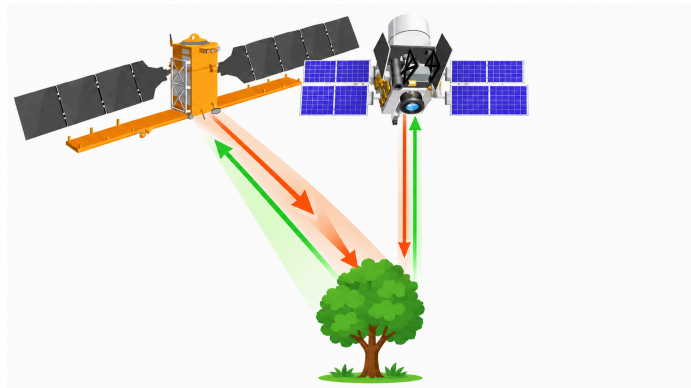
Passive Remote Sensing



Sunlight → reflected → sensed by the satellite

Example: optical satellites

Active Remote Sensing



Satellite emits energy → reflected → sensed

Examples: nadir lidar and side-looking radar

Earth Observation Satellites

Passive Remote Sensing - Examples from 07.02.2020

High Resolution

e.g., WorldView-3, PlanetScope, < 10 m

Medium Resolution 10m-30m

Sentinel-2, 13-band multispectral

Low Resolution

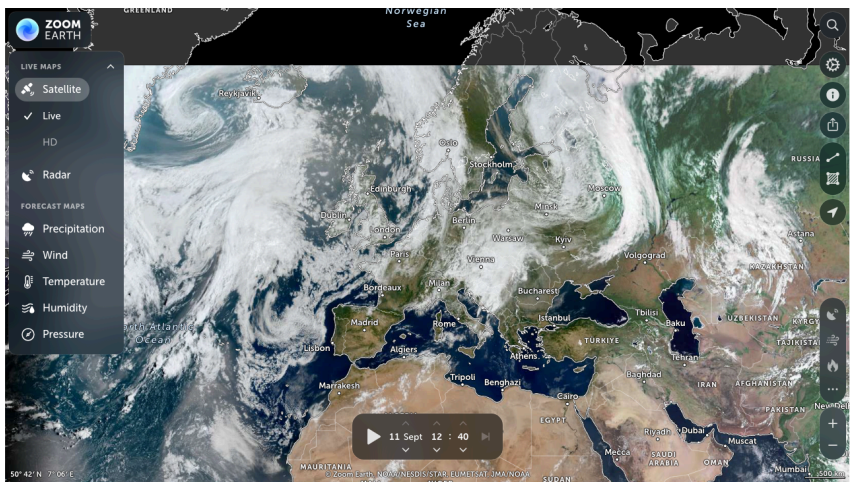
MODIS, 250m - 1km, 36-band multispectral

Weather and Meteorological Satellites

Passive Remote Sensing

METEOSAT — Geostationary Orbit (~36 000 km)

📍 Fixed view over Europe & Africa 🔭 500 m – 2 km 🕐 Every 10 min

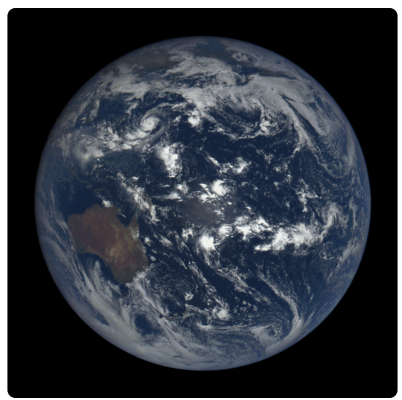


Click to open the live satellite view. Source: Zoom Earth.

DSCOVR EPIC — L1 Lagrange Point (~1.5M km)

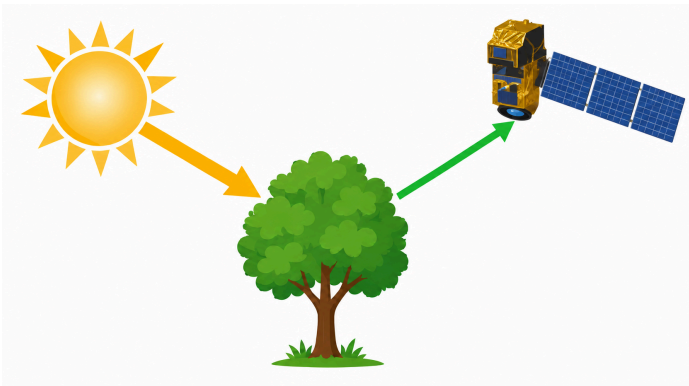
📍 Full sunlit Earth in every image

🔭 8 – 20 km · 🕐 Every ~2 h



DSCOVR EPIC — NASA / Wikimedia Commons. Public domain.

Takeaways — Passive Remote Sensing

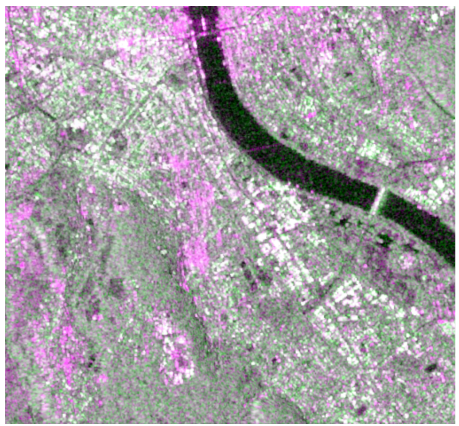


Sunlight → reflected → sensed by the satellite

Example: optical satellites

- Uses reflected sunlight by day or emitted thermal radiation day and night.
- Clouds and the atmosphere can obscure optical observations.
- Resolution trade-offs span space, spectrum, time, and radiometry.
- Higher spatial resolution does not inherently mean fewer bands.
- Very-high-resolution imagery is often commercial; coarser data are often open.
- Coarser sensors usually cover wider areas more frequently.
- Illumination and viewing geometry affect the signal.
- Natural-colour imagery is intuitive; reflectance requires atmospheric correction.

Active Remote Sensing - RADAR and LIDAR



Sentinel-1 SAR Raster

Side-looking radar processed as a pixel grid

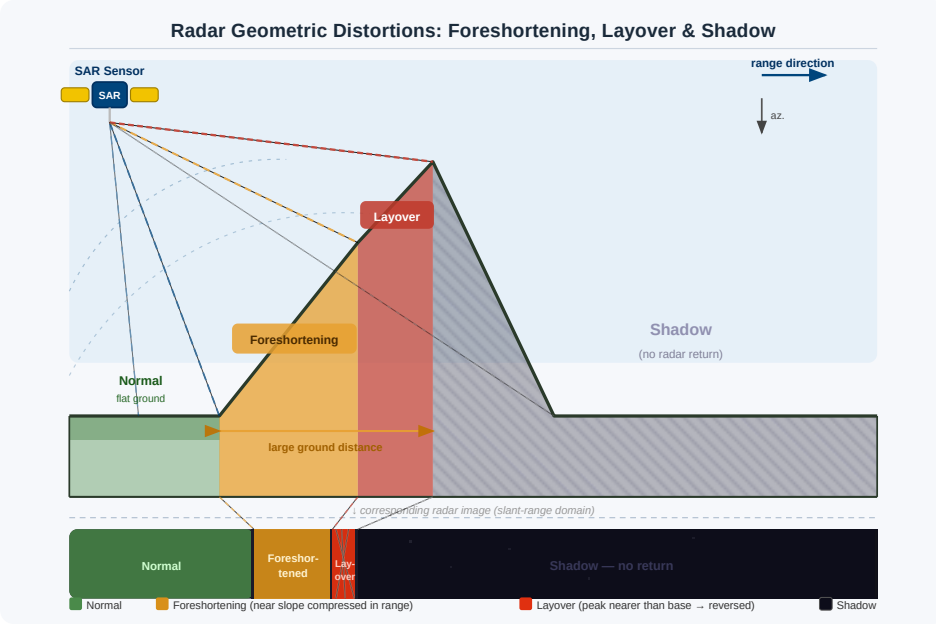
Radar Altimetry Tracks

Nadir measurements along point tracks

GEDI Lidar Footprints

Nadir samples along point tracks

Geometry of side-looking RADAR

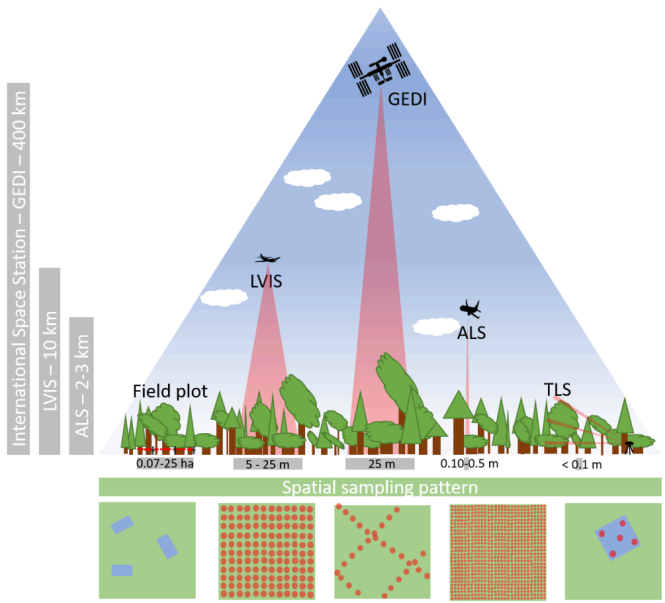


Example: TerraSAR-X

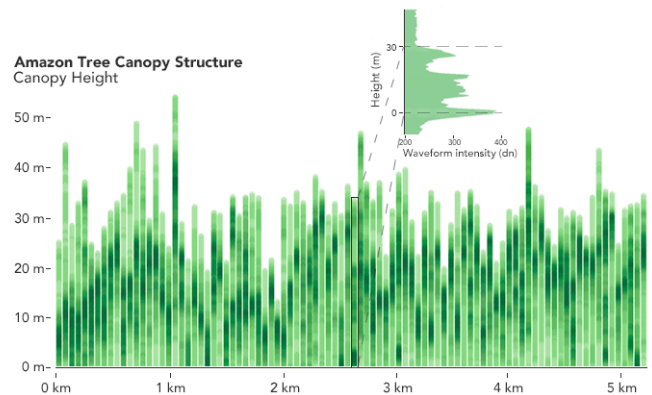
Figure uploaded by Richard Bamler.

LIDAR Backscatter for Vegetation Canopy

Sampling Geometry



Canopy Structure



Figures in both panels: NASA and University of Maryland.

Gravity Measurements from Two Satellites

GRACE Follow-On

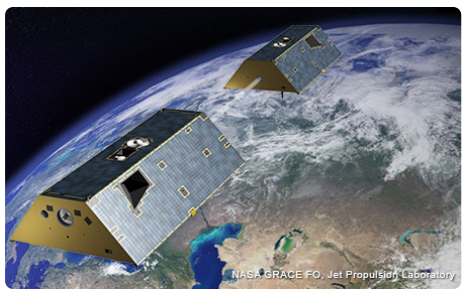
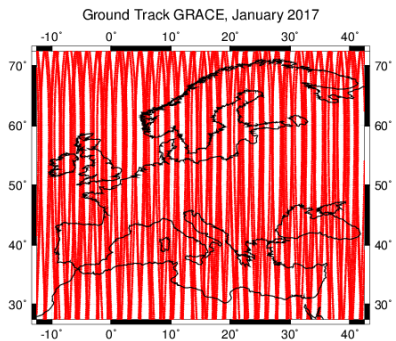


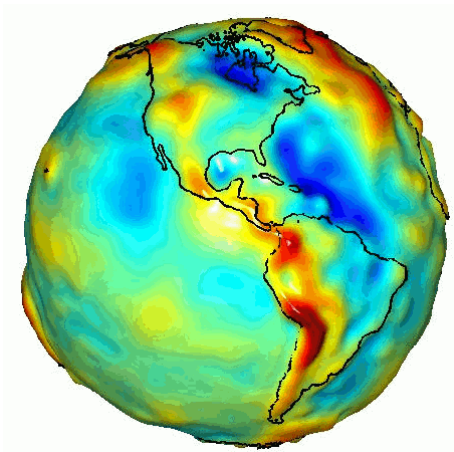
Figure: NASA/JPL.

Ground Track



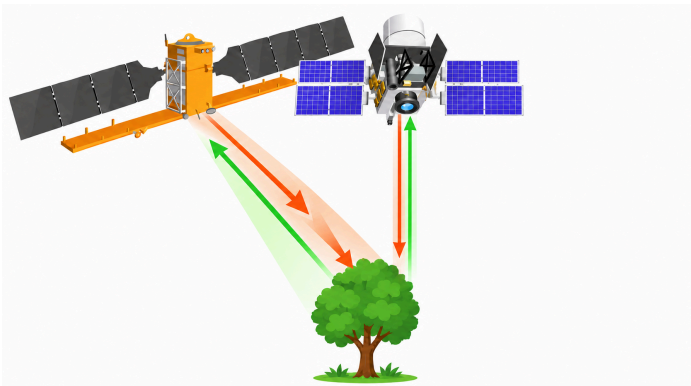
Ground Track GRACE, January 2017. Credit: Ch. Dahle, GFZ.

GRACE Geoid



Animation: Wikipedia; NASA/JPL/University of Texas Center for Space Research.

Takeaways — Active Remote Sensing

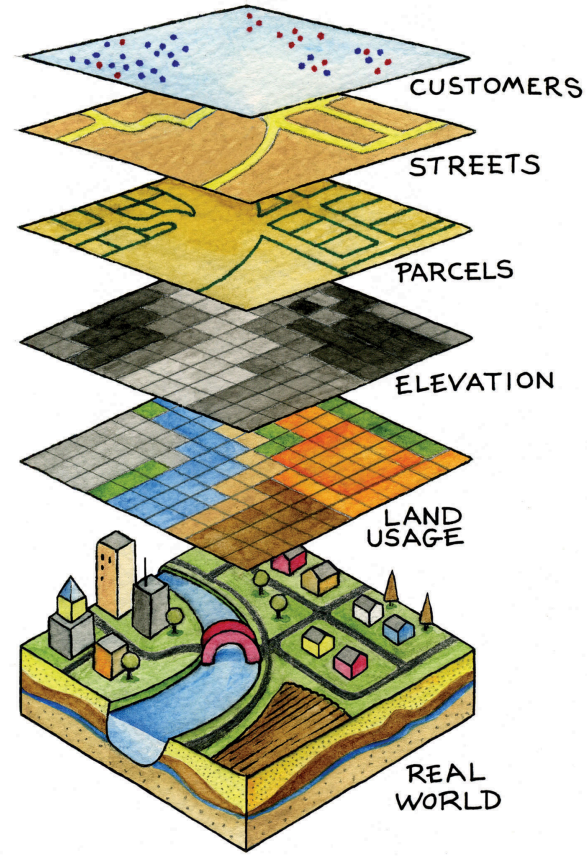


Sensor emits signal → return measured

Examples: SAR, LiDAR, radar altimetry

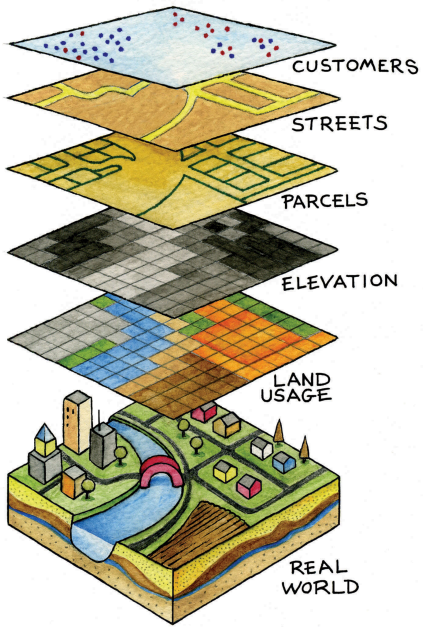
- Carries its own illumination — **independent of sunlight**, enabling imaging at night and in polar winter.
- **Penetrates clouds** and light rain (radar), overcoming a key limitation of optical sensors.
- Measures **physical surface properties** directly: backscatter, elevation, distance, or canopy structure.
- SAR captures **surface roughness, moisture, and deformation** through phase and amplitude.
- Radar altimetry delivers precise **surface height** along nadir tracks — not a full 2D image.
- LiDAR records **3D point clouds**: tree height, building geometry, terrain beneath vegetation.
- Shorter wavelengths (LiDAR) give finer detail; longer wavelengths (L-band radar) penetrate canopy and soil.

Geo-information Science



Campbell & Shin (2011), Figure 1.8.

Geo-information Science



Campbell & Shin (2011), Fig. 1.8

GIS organises geographic information as **stacked, queryable layers** — each layer aligns to the same coordinate reference system and carries thematic information about the world.

Vector layers

Points, lines, polygons + attribute table

Raster layers

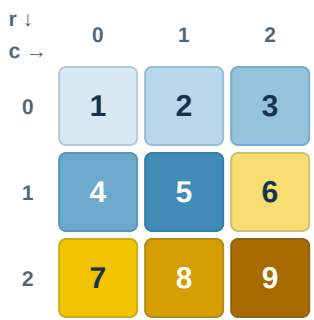
Grids of continuous values: elevation, imagery

Key data sources: OpenStreetMap, administrative databases, land use / land cover classifications, cadastral records, sensor networks.

Raster Georeference

Raster images are georeferenced by an affine transformation A that defines the top left coordinate with pixel sizes.

$$[East, North, 1] = A\ [Col, Row, 1]$$

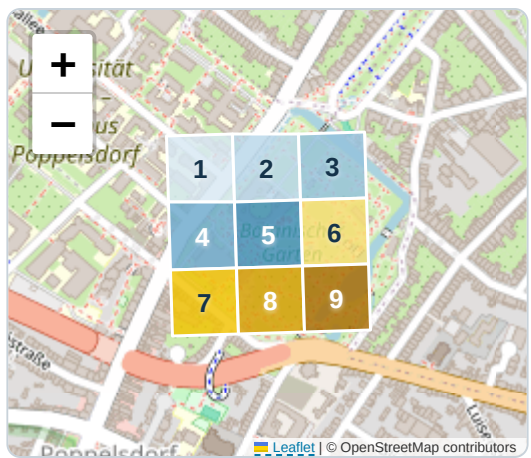


EPSG:32632 · WGS 84 / UTM zone 32N · metres

$$A = \begin{bmatrix} 100 & 0 & 365200 \\ 0 & -100 & 5621100 \\ 0 & 0 & 1 \end{bmatrix}$$

Apply transform

Initial transform applied



UTM coordinates are converted for the basemap display.

Task: Georeference this Image

EPSG:32632 · WGS 84 / UTM zone 32N · metres

$$A = \begin{bmatrix} & & \\ & & \\ 0 & 0 & 1 \end{bmatrix}$$

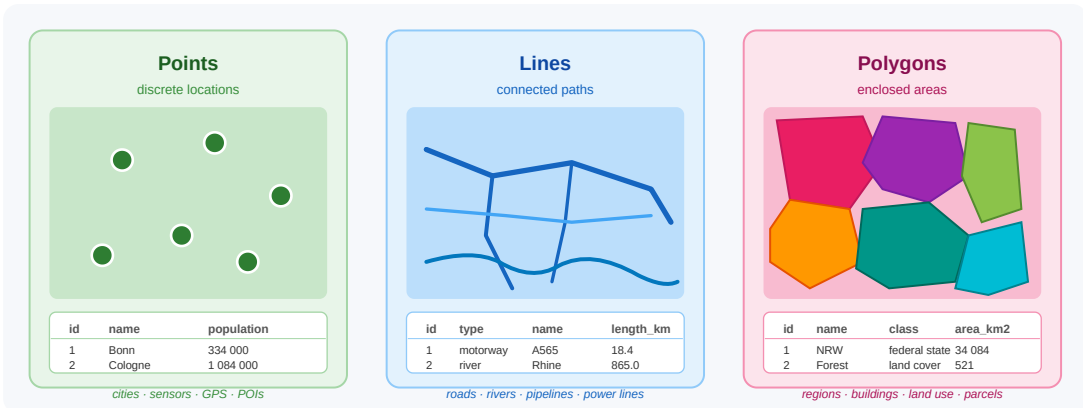
Transform

Fill all six entries, then transform

The **gdalinfo** output below provides metadata for this georeferenced image.

```
❯ gdalinfo Sentinel2_RGB_cloudfree_EPSG32632.tif
Driver: GTiff/GeoTIFF
Files: Sentinel2_RGB_cloudfree_EPSG32632.tif
Size is 47, 46
Coordinate System is:
PROJCRS["WGS 84 / UTM zone 32N",
    BASEGEOGCRS["WGS 84",
        DATUM["World Geodetic System 1984",
            ELLIPSOID["WGS 84",6378137,298.257223563,
                LENGTHUNIT["metre",1]]],
        PRIMEM["Greenwich",0,
            ANGLEUNIT["degree",0.0174532925199433]],
        ID["EPSG",4326]],
```

Vector Data — Geometry + Attributes



🟢 Points

Zero-dimensional locations: weather stations, cities, GPS waypoints, species occurrence records.

🔵 Lines

Ordered vertex sequences: roads, rivers, coastlines, pipelines, power lines, flight paths.

🔴 Polygons

Closed rings enclosing area: administrative units, building footprints, land-use patches, parcels.

Explore Vector Geometries

GeoJSON

106 features


```
{
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"name": "combined_osm",
"xy_coordinate_resolution": 0.000000,
"features": [
{ "type": "Feature", "id": "map – points_678973937",
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null, "is_in": null, "man_made": null, "name": "Zoologisches
Institut", "osm_id": "678973937", "other_tags":
"\"addr:city\"=>\"Bonn\",\"addr:country\"=>\"DE\",\"addr:house
number\"=>\"169\",\"addr:postcode\"=>\"53115\",\"addr:street\"
=>\"Meckenheimer Allee\",\"operator\"=>\"Universität
Bonn\",\"wheelchair\"=>\"limited\",\"wheelchair:description\"=
>\"Aufzug vorhanden, nur mit Schlüssel zugänglich\"", "place":
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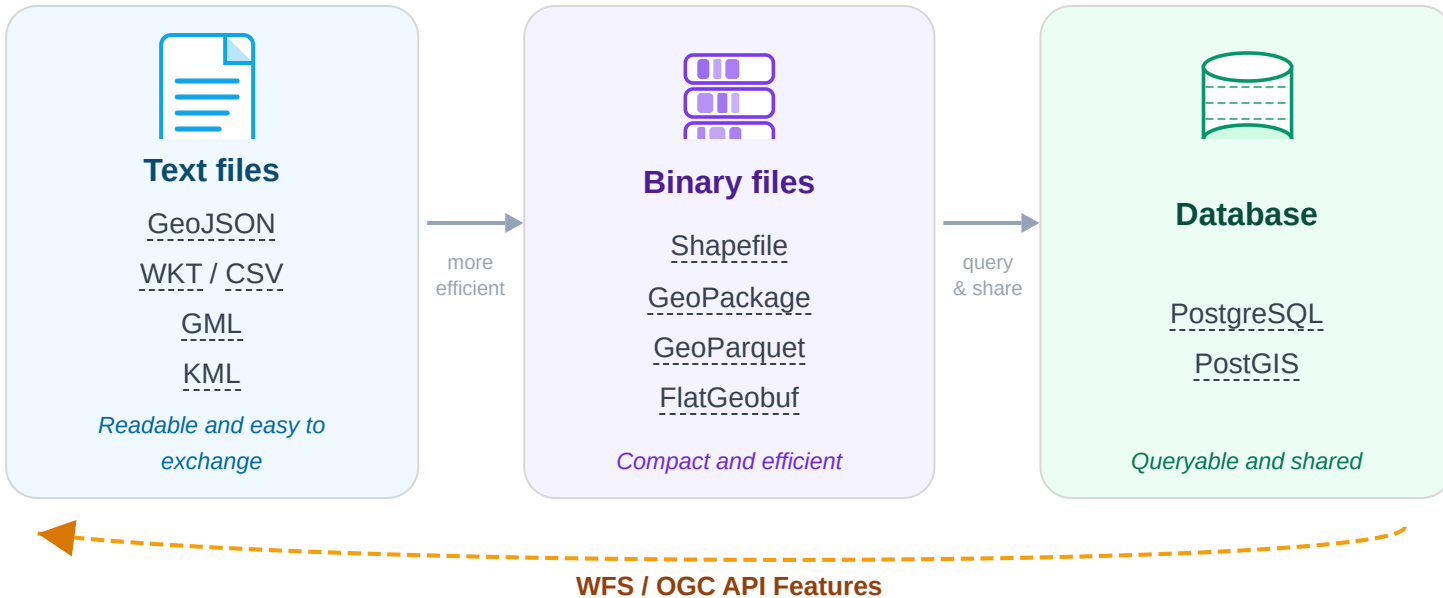
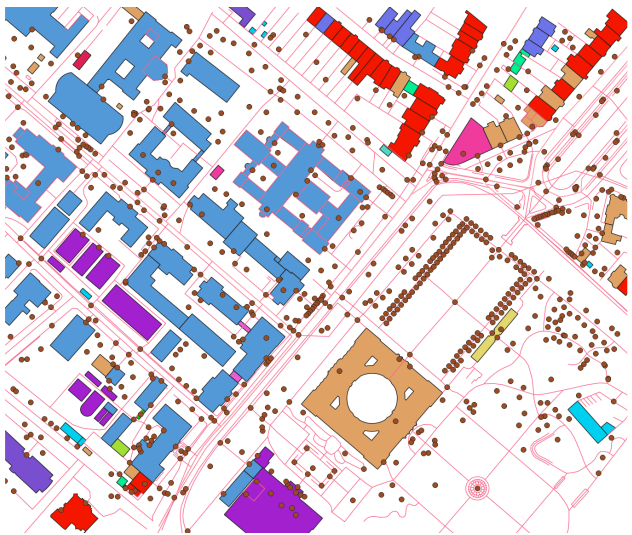
Leaflet map

Geospatial Vector Data Formats & Storage



OpenStreetMap — the Wikipedia of geodata

OSM Database



Points, lines, and polygons

Style and rasterize

→

OSM Background Map

Rendered raster tiles

Map data © OpenStreetMap contributors

Web Map Service (WMS)

```
https://ows.terrestris.de/osm/service?
SERVICE=WMS&VERSION=1.1.1&REQUEST=GetMap&LAYERS=OSM-
WMS&STYLES=&SRS=EPSG:25832&BBOX=363400,5620400,367000,5621600&WIDTH=900
&HEIGHT=300&FORMAT=image/png
```

Load map

BBOX = WEST, SOUTH, EAST, NORTH

Edit the request, then load the map

© OpenStreetMap contributors · WMS provided by terrestris

Web Feature Service (WFS)

```
https://www.wfs.nrw.de/geobasis/wfs_nw_alkis_vereinfacht?
SERVICE=WFS&VERSION=2.0.0&REQUEST=GetFeature&TYPENAMES=ave:GebaeudeBauwerk&COUNT=1000&SRSNAME=urn:ogc:def:crs:EPSG::4326&BBOX=50.719330278357,7.064421982400,50.730951027518,7.115835758151,urn:ogc:def:crs:EPSG::4326
```

Run WFS query

`BBOX` = WEST, SOUTH, EAST, NORTH

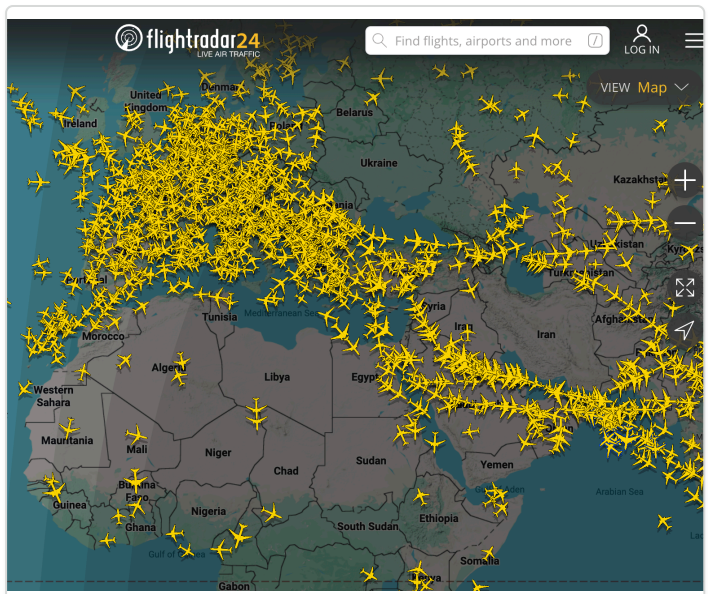
WMS: pixels · WFS: feature geometry

Edit the request, then run the WFS query

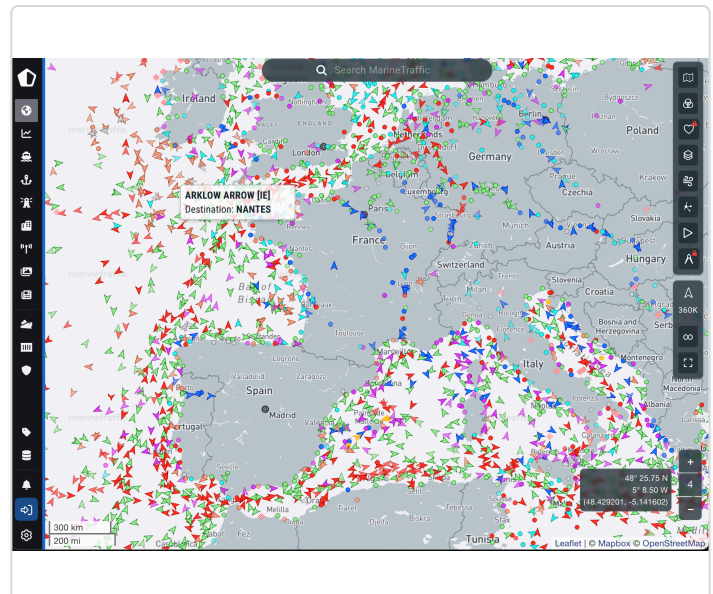
WFS data © Geobasis NRW

WFS/WMS in Practice

Flightradar24



MarineTraffic



Administrative Boundaries

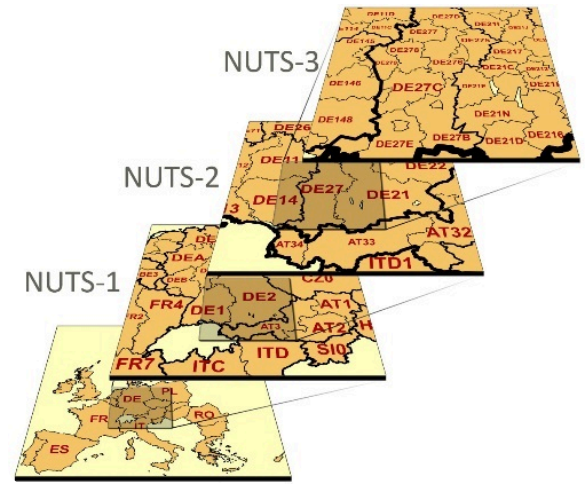
GADM — Global Administrative Areas

Germany · 6 nesting levels · global coverage

Freely available polygons for every country; widely used in ecological & socioeconomic modelling.

Gringer / Wikipedia · CC BY-SA

NUTS — EU Territorial Statistics Units



NUTS-1 → NUTS-2 → NUTS-3 · three nesting levels

EU standard: NUTS-1 (federal states), NUTS-2 (Regierungsbezirke), NUTS-3 (districts). Reference frame for Eurostat statistics.

© Eurostat · ec.europa.eu/eurostat

Land Use / Land Cover (LULC) Databases

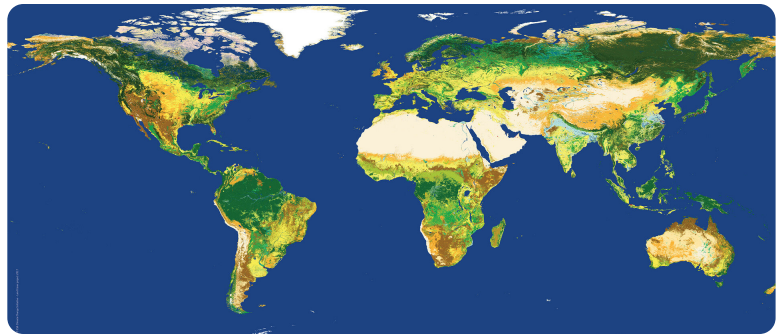
Land use describes human activity (agriculture, urban, forestry). **Land cover** describes the physical surface (vegetation, bare soil, water).

ESA WorldCover 2020–2021
10 m global · 11 classes · Sentinel-1 & -2

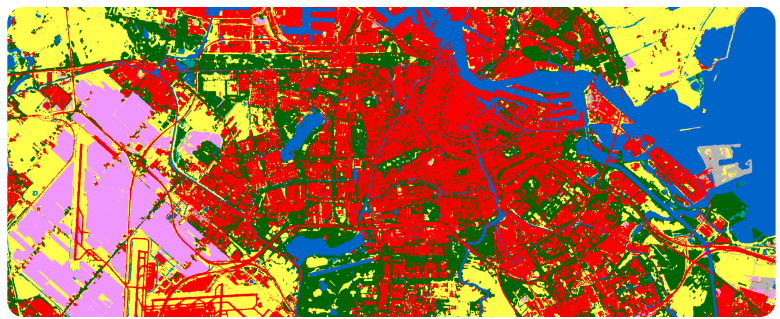
CORINE Land Cover (CLC) EU, since 1990
100 m · 44 classes · updated every 6 years

GlobeLand30 Global
30 m · 10 classes · 2000 / 2010 / 2020 epochs

Dynamic World (Google) Near-real-time
10 m · 9 classes · probabilistic per-scene output



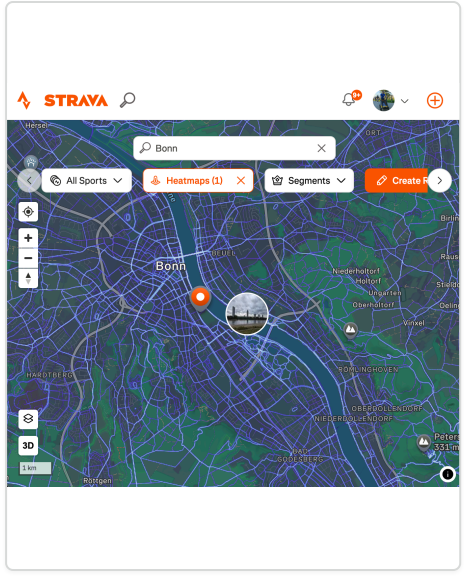
Global land cover map 2015 · © ESA / ESA Climate Change Initiative



ESA WorldCover 10 m — Amsterdam · © Copernicus / ESA

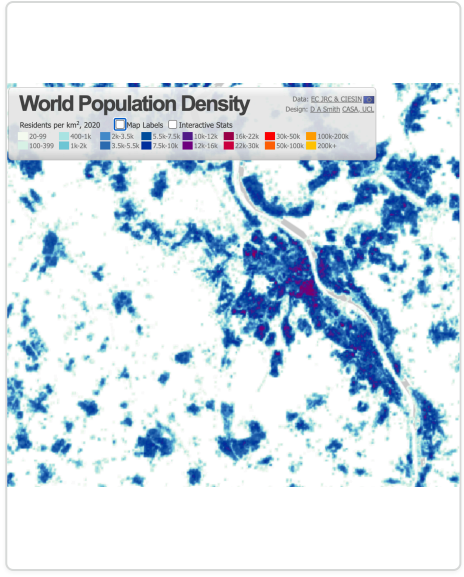
Further GIS Examples

Strava Heat Map



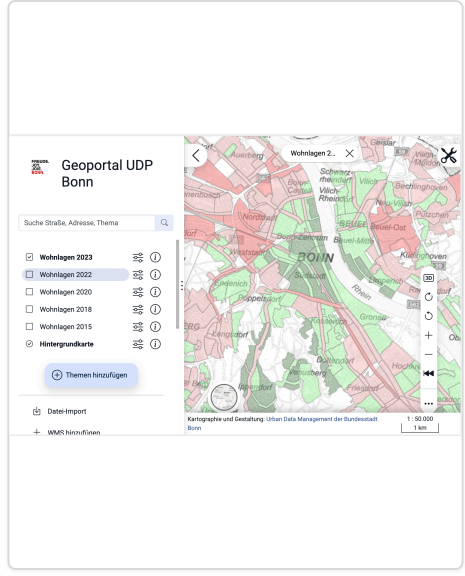
Screenshot: Strava

Population Density



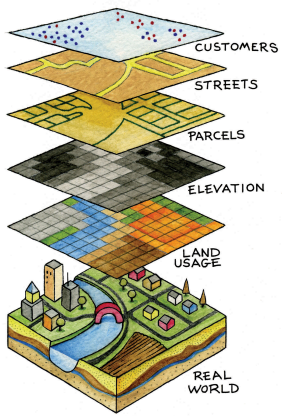
Screenshot: Luminocity3D

Quality of Living (Wohnlagen)



Screenshot: Gutachterausschuss Bonn

Takeaways - Geo-information Science



Real world → thematic layers → analysis

Examples: OSM, CORINE, GADM, cadastres

- GIS organises the world as **thematic layers** aligned to a shared coordinate reference system.
- Vector data combines **geometry** (points, lines, polygons) with an **attribute table** — enabling spatial and semantic queries.
- OpenStreetMap is the world's largest **open vector database**: roads, buildings, land use, and POIs contributed by millions.
- LULC databases (**ESA WorldCover, CORINE**) map land cover classes globally using satellite imagery — raster products with thematic labels.
- Administrative boundaries (**GADM, NUTS**) are the standard aggregation frame for statistics, policy, and spatial cross-validation.
- **Spatial joins** link vector layers: assign points to regions, aggregate raster values within polygons.

Models and Re-analysis Products

From Observations to Modelled Fields

Combine observations with physical models to estimate how Earth's systems evolve.

Observations → **Model + data assimilation** → **Estimated fields**

- Satellites, stations, buoys, and profiles provide observations.
- Physical models describe evolution; data assimilation updates the model state.
- Outputs describe conditions across space, time, and height or depth.

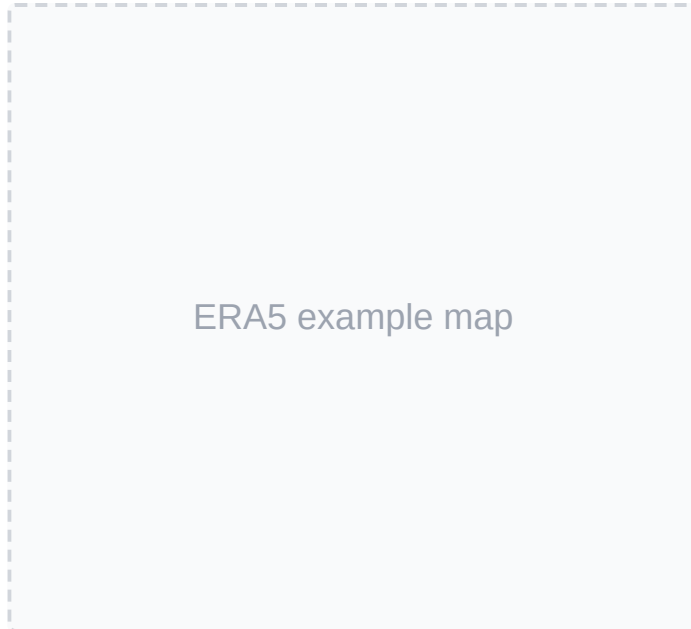
Analysis, Forecast, Hindcast, Reanalysis, Projection

Product	Purpose
Analysis	Estimate the state at a given time using a model and available observations.
Forecast	Predict subsequent conditions from an initialized state.
Hindcast	Run a model for a past period; assimilation depends on the experiment.
Reanalysis	Reconstruct the past by assimilating archived observations with a consistent modelling system.
Climate projection	Simulate possible future climate under specified forcing scenarios.

ERA5 — Reconstructing the Atmosphere

- Global atmospheric reanalysis combining observations and a numerical model.
- Temperature, wind, pressure, precipitation, and other variables.
- Hourly fields at the surface and multiple atmospheric levels.

Example: At a location and time, retrieve estimated temperature and wind.



Source: ECMWF ERA5

HYCOM — Modelling the Ocean

- Ocean dynamics represented across horizontal locations and vertical layers.
- Temperature, salinity, sea level, and current velocity.
- HYCOM-based products include analyses, forecasts, and historical simulations.

Example: At a location, depth, and time, retrieve temperature and current velocity.

HYCOM example map

Source: HYCOM

Model Outputs as Geospatial Data

```
value = f(location, height/depth, time, variable)
```

- Usually multidimensional gridded fields; common formats include NetCDF, GRIB, and Zarr.
- Select a time and level for a map, or a location for a time series.
- Other examples: ERA5-Land (land), CAMS (atmospheric composition), GloFAS (rivers), CMIP6 (climate projections).

Takeaways — Models and Re-analysis Products

- Model products provide estimates of environmental conditions across space and time.
- Analysis, forecast, hindcast, reanalysis, and projection answer different questions.
- Check the product, time definition, resolution, units, and uncertainty before using a value.

A complete grid does not imply perfect knowledge.